Investigating the Impact of Generative AI Tutoring Styles on Student Knowledge Acquisition and Confidence Calibration

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1 Introduction

The rise of Generative AI (GenAI), along with all of its capabilities, has become a massive conversation in the education industry [1]. Ever since the release of ChatGPT by OpenAI in November 2022, the education landscape has evolved tremendously [1]. While schools initially took a restrictive stance by completely banning the use of AI, the trend has shifted to embracing and integrating GenAI into their pedagogy [2]. One key factor that drives this transition is GenAI's ability to provide personalized learning and customized feedback based on the unique needs of each individual student, which human tutors struggle to provide at scale [3]. The presence of GenAI enables students to focus on higher-order skills such as critical thinking and creativity rather than repetitive tasks [4, 5]. Moreover, due to today's rapidly evolving professional landscape, the adoption of GenAI in higher education has become a necessity rather than an option [6]. Hence, institutions need to create an environment where AI can be used to enhance human intelligence rather than replace it [1, 5].

However, the adoption of GenAI in higher education comes with a significant set of challenges. As previously mentioned, GenAI can be useful in taking over repetitive tasks [7], but this also leads to "cognitive offloading" where students outsource critical cognitive tasks to the AI [7]. The pattern of reliance starts from using AI as an information verification tool to slowly becoming a content delegation tool, causing students to blindly rely on AI to generate the final output [8]. It becomes even more concerning as AI creates a "productivity paradox", where despite students experiencing increased productivity, they lack depth in critical analysis [9]. For instance, due to convenience and speed that AI summaries provide, students tend to prioritize these summaries over verifying the underlying sources [10]. Despite the faster drafting process, the final result ended up being lower quality, with generic essays and a lack of critical analysis [1, 9]. Moreover, this matter is further supported by the so-called "death of essays", a phenomenon that has risen to prominence since COVID-19 due to the increase in take-home exams and students utilizing AI to cheat and plagiarize [1]. This raises the importance of AI literacy. Studies show that students with higher AI literacy tend to take a collaborative approach by planning and evaluating AI outputs, which consequently leads to better outcomes compared to students with lower AI literacy, who tend to over-rely on AI just to get the job done [11]. Moreover, over-reliance on AI negatively leads to the decline of face-to-face interactions, which can cause implications on student's interpersonal skills and emotional intelligence. Consequently, this increases the risk of social anxiety and isolation [12].

As previously mentioned, over-reliance on AI is a significant concern that must be addressed. AI tends to generate text with a convincing tone, giving an illusion that the output is always true, causing users to overlook hallucinations and in the end hallucinating themselves [1, 10]. To combat this, students are required to possess foundational knowledge before adopting the tool [1, 13]. This ensures they are able to critically validate the AI's information, as the human in the loop is responsible for the final output produced [1, 14]. Therefore, students must be taught to verify AI outputs [15]. On the other hand, educators are encouraged to create critical thinking assessments that can't be replicated by AI, such as video presentations or pitches [15, 1].

This gap creates a strong opportunity for designing how students should interact with AI. As previously mentioned, AI tutoring styles are not static [3]. The characteristics of the AI's interaction can be engineered using prompts (prompt engineering) and guardrails [16]. In other words, poor learning outcomes are not solely based on the AI's accuracy and performance, but also an issue with the interaction design itself [17, 18].

To address the illusion of truth, AI tools must be designed with guardrails that encourage cognitive engagement rather than handing out direct answers [19]. In the context of learning, the design of the AI interaction itself is more important than its capabilities [18]. Evidence shows that the use of scaffolding techniques, such as guiding hints and follow-up questions, are able to significantly improve students' learning outcomes [18]. AI chatbots with these guardrails are referred to as Socratic chatbots, which are designed to prevent giving out direct answers to encourage self-reflection and deep understanding of the topic [13, 20, 21]. When compared to default GenAI models, Socratic chatbots are able to produce 79% more learning improvements than the default models (127% vs. 48%) during practice assessment [18]. Additionally, unlike the default model where students experience learning loss when AI is removed, students did not experience any learning loss when the Socratic chatbot is removed [18]. This indicates that the default AI interaction may have short-term benefits, but confidence outpaces actual understanding, creating an illusion of competence [18]. Although the Socratic tutoring style works in many contexts, it is not universally superior [22]. In the context of coding, a study comparing the effects of Socratic AI versus normal AI shows that Socratic AI produces a higher error ratio and has no evidence of improving long-term outcomes [22]. Furthermore, learners with lower AI literacy tend to find scaffolding bots frustrating, which further amplifies the need for alternative tutoring styles [13]. This underscores the need to look beyond Socratic tutoring. It is important to explore other tutoring styles that could effectively improve both learning and confidence calibration.

Yet most literature rarely compares alternative interaction designs within the same study. Despite growing evidence on AI-supported learning, most studies evaluate one tutoring style at a time. As a result, the field still lacks clear comparative evidence on which tutoring style best results in balanced learning outcomes as well as psychological

learning benefits. In practice, GenAI can be deployed with a spectrum of tutoring styles using prompt engineering and guardrails [16]. Beyond Socratic approaches, evidence shows that reverse tutoring can be effective in boosting cognitive engagement and knowledge mastery [23, 24, 17, 25]. This pedagogical approach embodies the theory of learning-by-teaching by using recursive feedback that observes how a tutee applies what has been taught [23]. This approach is similar to designs where the AI takes the role of a student, prompting the human learner to teach the AI, thereby reinforcing the learner's own understanding [15]. Studies show that reverse tutoring creates the protégé effect where students learn harder when they know they must teach [23, 25]. There are two mechanisms that align with reverse tutoring [24]:

1. Generative processing: Students are encouraged to inherently organize and integrate their knowledge.
2. Metacognitive monitoring: Ability to detect explanation gaps, triggering self-correction.

Moreover, these interactions helps drive the learners to dig deeper into the topic which leads to a more flexible mastery of the topic [24, 17].

Hence, in addition to the Socratic tutoring style, this study introduces the reverse tutoring technique. This study compares the three tutoring styles (direct-answer, Socratic, and reverse tutoring) within a single comparative framework, while assessing both cognitive and psychological outcomes. In particular, reverse tutoring, which is grounded in learning-by-teaching theory, despite showing clear effectiveness in improving learning outcomes, remains underexplored in the GenAI context. Therefore, this study aims to fill that gap by asking how different tutoring styles compare in their effects on learning performance, knowledge acquisition, self-efficacy, and confidence calibration. This combined evaluation is important, as prior work shows that students can appear more productive with AI, but could lose their learning once the tool is removed [9, 18]. Accordingly, this study aims to determine which tutoring style best supports durable learning rather than short-term productivity.

2 Literature Review

2.1 Generative AI in Education

It cannot be denied that AI has revolutionized the way the education industry operates [6]. In fact, AI has been implemented within the education industry for decades [21]. Introduced by Jaime R. Carbonell in 1970, the SCHOLAR tutor system was the first intelligent tutoring system (ITS) [26]. By utilizing a semantic network of concepts and facts, the system was able to evaluate students' knowledge of geography, creating a two-way interaction with the student [26]. Over the past decade, specialized AI tools such as Wolfram Alpha and Geogebra have enhanced STEM learning by increasing interactivity, while tools like Grammarly have assisted students in writing [21]. Although these tools were effective, they lack the conversational capabilities that current GenAI models possess. Driven by natural language processing (NLP), these large language models (LLMs) are able to hold sustained dialogue and have since changed the way humans interact with information [21, 1].

The presence of GenAI has not only changed the way students and teachers interact, but also the entire curriculum at all levels of education [15, 2, 27]. From initially banning the use of AI due to concerns about cheating and plagiarism, institutions are now embracing and integrating AI into their pedagogy [2, 6]. This shift is supported by GenAI's ability to deliver an adaptive and personalized learning experience for students [3]. For a long time, education has been challenged in providing for the unique needs of each individual student [21]. Evidence shows that students experience an increase in their learning outcomes when being tutored in a one-on-one setting [28]. However, the cost of providing human one-on-one tutoring is high in both financial and human resources, especially as teachers are also burdened with administrative responsibilities [28, 15]. This is where GenAI comes into play by providing a scalable and cost-effective one-on-one tutor compared to traditional human tutors, which cost $3,500–$4,300 per student per year [28]. Hence, it lowers the entry barrier for complex subjects by providing examples and analogies that cater to the user's understanding and interests [2, 29]. For example, in the context of vocabulary learning, the AI's output can be personalized based on the user's hobbies, such as cooking, sci-fi, or sports, turning a dry learning experience into a more engaging and interactive one [29, 1].

Despite the immense benefits that GenAI provides, when used without proper guidance, it can cause harm to long-term learning outcomes [18]. Therefore, it is critical for educators to standardize GenAI's use in education. In order to correctly adopt GenAI in their pedagogy, educators are encouraged to follow UNESCO's AI Competency Framework, which acts as a guide for AI use and misuse [30]. The framework highlights several key dimensions essential to effective AI implementation: students must develop foundational knowledge for overseeing and validating the final output [30, 14]. Thus, institutions must create an environment where AI enhances human intelligence rather than replaces it, fostering higher-order thinking and creativity [5, 1]. Moreover, GenAI should be prevented if it removes the opportunity for students to develop cognitive abilities or social skills [1].

However, in practice, GenAI is still being used incorrectly, leading to short-term productivity but long-term learning loss, known as the "productivity paradox" [9, 18]. This phenomenon occurs because students tend to offload cognitive tasks to the AI, such as planning, organizing, and evaluating [7]. Evidence shows that most students prioritize the convenience and speed of AI summaries over verifying the underlying sources [10]. Hence, students who over-rely on AI are found to produce generic and lower-quality work [1]. Consequently, the excessive use of ChatGPT reduces their academic performance and memory [31]. This shows that GenAI cannot yet be supportive of self-regulated learning unless it is pedagogically engineered to provide hints and scaffolds rather than direct answers [7]. To mitigate the misuse of GenAI, educators are encouraged to provide assessments that require students to document their process in reaching their final output (e.g., oral examination or AI-output documentation) [27, 1]. Furthermore, institutions are encouraged to use GenAI to foster higher-order skills and creativity, combining traditional pedagogy with AI-assisted exploration [5, 2]. Thus, it is important to design a human-centric model where GenAI acts as a cognitive partner rather than a replacement [5].

2.1.1 Socratic AI versus Direct AI

As previously mentioned, without proper guardrails, GenAI outputs direct answers, posing the risk of cognitive offloading [7, 5]. To prevent this, Socratic chatbots are designed using prompt engineering and guardrails to encourage cognitive engagement by offering hints and guidance rather than direct answers [16, 20]. As the name suggests, the design of Socratic AI chatbots originated from the Socratic teaching method [21]. The method itself was originally designed to doubt all assumptions using questions and dialogues, forcing students to construct their own knowledge [32]. This technique has proven to be effective in improving students' learning outcomes, increasing their learning gains by 59% in the context of coding comprehension [33]. In addition, a comparative study of Socratic AI versus direct AI shows that Socratic AI is able to improve learning gains by 127% with no learning loss when the AI is removed [18]. In comparison, the default model shows a learning loss of 17% when the AI is removed [18]. Open-ended questions and reflective prompts have been proven to be more useful in promoting self-regulated learning, as agreed by 85% of educators [34]. This is due to the back-and-forth interaction that encourages students to verbalize their reasoning and correct any misconceptions, in comparison to the default model that gives direct solutions and deprives students of cognitive effort [35]. Thus, it can be acknowledged that the design of the AI interaction is more important than the model's capabilities in improving learning outcomes [18].

Additionally, the convincing tone of default AI models can lead to users believing that every response is true, despite the inaccuracies caused by hallucinations [1, 36]. This "illusion of truth" can be harmful, as it leads users to overlook these hallucinations, causing errors in their final output [1, 10]. Subsequently, users' confidence tends to align with the AI-expressed confidence, further imposing the danger of this illusion [37]. Moreover, the authoritative tone of the AI leads students to become over-reliant on the AI, which has been proven to limit creativity and critical thinking, as well as social and emotional development [11, 37]. Furthermore, being over-reliant on the AI leads to a lack of face-to-face interaction, which negatively impacts students' interpersonal skills and leads to a higher risk of social anxiety and isolation [12].

To mitigate this, Lo et al. proposed the CLEAR framework, which stands for Critical, Reflective, Adaptive, Explainable, and Responsible [14]. This framework encourages students to critically analyze the AI's output, reflect on their own learning, adapt to feedback, explain the reasoning behind their answers, and be responsible for the final output [14]. However, to be able to apply this framework, students must possess foundational knowledge of the topic as well as AI literacy to critically evaluate the AI's output [14]. Simply flagging whether the content is fake is not effective, as students must develop internal critical thinking to succeed [38]. A study comparing traditional study methods, studying with direct AI, and studying with Socratic AI approaches found that users who studied with direct AI showed a learning loss of 17% when the AI was removed, while users who studied with the traditional method and Socratic AI showed no learning loss [18]. This further proves that Socratic AI chatbots go beyond fact-checking, as they allow students to recognize cognitive biases and build emotional triggers, encouraging them to become more cognitively active [38].

Despite being effective, these Socratic chatbots are not a one-size-fits-all solution. Learners with low AI literacy tend to find these scaffolding bots frustrating [13]. When compared to direct AI in the context of coding, students who were assigned the Socratic AI actually performed worse, with lower efficiency and a higher error ratio [22]. Additionally, no significant long-term learning outcome was found, meaning the domain and task matter [22]. Hence, in order to achieve productive human-AI collaboration, students must possess high AI literacy as a prerequisite [13, 11]. Studies show that students with higher AI literacy are able to evaluate and steer the AI to receive deeper insights, while lower-literacy learners revert to simple queries [13, 11]. Therefore, it is necessary for educators to provide prompt literacy to students, fostering UNESCO's AI Competency Framework that envisions students as AI co-creators [5, 30]. As previously acknowledged, each student has their own unique needs [21]. Therefore, research on designing other human-centric models should also be pursued to cater to different learning styles and needs [5].

2.2 Various AI Learning Strategies

Beyond Socratic AI chatbots, students are able to engage with AI through a variety of interaction styles, each offering their own benefits and risks, as shown in Table 1 [15].

AI Style	Description	Benefit	Risk
Mentor	Provides specific feedback and guidance on the student's work	The frequent feedback allows students to reflect even if not all advice is taken	May encourage over-reliance, where students do not critically evaluate the AI's feedback
Tutor	Offers direct instruction and answers	Effective for quick acquisition of knowledge	The uneven knowledge base could cause hallucinations
Coach	Engages with students metacognitively (thinking about their own thinking) by prompting reflections	Builds self-regulation and planning skills	Incorrect advice may lead to misguidance; the interaction style may not suit every student
Teammate	Collaborates with students to solve tasks by playing a devil's advocate role to challenge the student's assumptions	Provides alternate perspectives and fosters collaborative learning	May provide errors that misguide the student's understanding
Student	Reverses the role; the student teaches the AI	Promotes learning-by-teaching, which can uncover knowledge gaps	The AI's argumentative behavior may derail the benefits of this approach
Simulator	Provides realistic role-play scenarios to practice the student's skills	Practical and hands-on knowledge application aids in transfer of learning	The story element may distract from the core concept being taught
Tool	AI to extend the learner's capabilities by being a general-purpose utility	More work can be done within the same time frame	Risk of outsourcing the thinking process to the AI

Table 1: Comparison of various AI tutoring styles, their descriptions, benefits, and risks [15].

Additionally, the ICAP framework [39] provides a clear categorization of different levels of cognitive engagement:

1. Interactive: Learning by collaboration, where both parties contribute by building on each other's ideas (e.g., debate).
2. Constructive: Generating new information beyond the materials provided (e.g., mind mapping).
3. Active: Learning using physical manipulation without generating anything new (e.g., highlighting).
4. Passive: Receiving information without any physical engagement (e.g., listening to a lecture).

The ICAP framework has shown that as students move higher in the hierarchy from passive to interactive, learners experience an increase in learning due to the knowledge-change processes [39]. Drawing upon the different ways students can engage with AI and its benefits, as well as the validity of the ICAP framework, this study proposes to compare three different AI tutoring interactions: direct-answer, Socratic, and reverse-tutoring. The reverse-tutoring approach is chosen due to its potential in fostering active learning, which has been proven to increase learning outcomes [39].

2.2.1 Reverse-tutoring (The Protégé Effect)

"While we teach, we learn" is a famous quote by the Roman philosopher Seneca [25]. Beyond the quote, decades of research have shown that learning-by-teaching has proven effective in improving one's knowledge of a topic [25, 40]. The beneficial learning outcomes of learning-by-teaching are driven by socio-psychological mechanisms whereby being assigned a teacher role increases students' agency, consequently boosting their cognitive engagement and outcomes [25, 23]. This phenomenon is known as the "protégé effect," where students tend to learn more as they know they must teach [23]. The phenomenon is further supported by Fiorella and Logan's study, which found that the process of reverse-tutoring involved two mechanisms: generative processing and metacognitive monitoring [24].

- **Generative processing:** Learners generate new information beyond the materials provided and draw relationships between them. This process is driven by the learner's own understanding and knowledge.
- **Metacognitive monitoring:** As learners teach the material, they constantly monitor their own understanding, detecting explanation gaps.

The combination of these two mechanisms shifts learners' knowledge from surface-level understanding to a flexible mastery of the material [24]. These processes are aligned with the interactive level of engagement from the ICAP framework, as students are able to reflect on their own understanding and build upon knowledge gaps as they teach [39, 24]. Additionally, a comparative study by Kobayashi found that students who were explicitly told they were going to teach performed twice as well as those who were not told [40]. Moreover, students who later actually taught performed even better than those in preparation-only conditions, indicating that the act of communicating knowledge possesses a unique cognitive benefit beyond preparation alone [40].

One of the most frequently cited teachable agents (TA) is Betty's Brain (2007), a pedagogical agent built to foster reasoning for middle school students in the subject of biology [41]. Dialogue-based interaction was not possible in this era due to the lack of advancement in NLP [41]. Instead, students interact by building causal maps that act as the "brain" of the agent [41]. The study shows that the presence of the TA was able to absorb the blame for errors, allowing students to acknowledge their mistakes while protecting their ego [41]. Additionally, the TA's presence was incredibly beneficial for low-achieving students, who were able to score equal to the high-achieving students on the hardest questions [42, 41]. Moreover, the benefits of TAs persisted even after the technology was removed, drawing similarities to Socratic AI, where learning outcomes are retained even after the AI is removed [43, 18].

Putting the learning-by-teaching theory into the context of GenAI, we reverse the roles of the student and the AI by having the student teach the AI as a naive tutee [15]. This can be done by designing the personality of the chatbot to be curious and open-minded [17]. By doing so, students are driven to give higher-order explanations and engage in more interactive exchanges [17]. The AI should act as a recursive feedback tutor that observes what the student has taught it [23]. A study by Jin et al. developed a TA of their own called AlgoBo, which acts as a struggling novice in the subject of coding [44]. AlgoBo is prompt-engineered to have strict knowledge of only documents within the chat, and acts as either a help-receiver or an active "questioner" that asks why and how questions to the student [44]. This strategy encourages students to think of edge cases and misconceptions, just as denoted by the protégé effect [24, 44]. Moreover, despite prompting students to produce deeper conversations, there was no increase in self-reported cognitive load [44].

It is important to note that the personality of these AIs matters [17]. If the AI fails to emphasize curiosity by showing agreeable and extroverted behavior, it may lead to shallow and repetitive exchanges [17]. Even though it may lead to an increase in interaction volume, the interaction is not driven by higher-order engagement [17]. Therefore, reverse-tutoring is not a one-size-fits-all solution, and it depends on the TA's persona design [17].

2.3 Confidence Calibration

Confidence calibration is defined as how well a person's subjective sense of certainty matches actual correctness [45]. In the context of human-AI interaction, confidence calibration usually refers to how much a user trusts the AI's output compared to how accurate the output is [45]. As previously mentioned, the illusion of truth—where modern LLMs can appear highly confident even when they output incorrect information—poses a significant risk of causing miscalibration [46, 1]. A study by Zhang et al. shows that users tend to rely more on the AI when the AI acts more confidently, causing users to overlook the AI's hallucinations [47]. This is especially dangerous for users with low AI literacy, as they might take in information without critically evaluating it, and eventually become over-reliant on the AI [11]. Furthermore, when miscalibration occurs, users tend to blame themselves for the mistake, destroying their self-efficacy, and eventually lean harder on the AI [45].

A consistent pattern across these studies is that calibration improves when the system shows uncertainty (e.g., asking the user to double-check) and when the human themselves can complement the AI's output [48, 46]. This uncertainty friction leads users to fact-check with their own knowledge or external sources, resulting in a deeper and more accurate final output [46]. Thus, designing an AI system that fosters uncertainty and prompts the user to fact-check with their own knowledge is crucial for achieving better calibration [46]. Additionally, literacy plays a crucial role in the interaction, as users with low literacy may find the uncertainty friction overwhelming [13]. Hence, AI usage education, such as the CLEAR framework [14], is crucial in making sure users are able to understand the uncertainty and fact-check with their own knowledge.

In the scope of this study, the different designs of AI tutoring styles will be compared in terms of their calibration performance to determine which tutoring style is most effective in fostering better calibration. The default direct-answer AI risks strong confidence inflation due to the lack of uncertainty friction, leading to overreliance on the AI's output [18]. Socratic AI may improve calibration by encouraging self-explanation and error checking, but may raise frustration

[13, 18]. Reverse-tutoring can surface knowledge gaps and reduce the illusion of competence through metacognitive monitoring [24]. Finally, the interaction between calibration and self-efficacy is important to note, as it has been proven to be a vicious cycle where bad AI advice can reduce the user's self-efficacy, which can then increase future dependence on the AI [45]. Following prior work, the use of multiple assessments (pre-test, AI-assisted practice, and post-test without AI) can be used to distinguish short-term task completion from retained knowledge and well-calibrated confidence [18].

3 Research Question

The main research question in this study is: How do (A) different tutoring styles influence the effectiveness of generative AI tutoring; specifically, how does it affect (B) student learning performance, (C) self-efficacy, and (D) confidence calibration under conditions of varying perceived task difficulty?

The tutoring style factors examined in this research are *(A1) direct-answer AI tutoring, (A2) Socratic AI tutoring, and (A3) reverse-tutoring (protégé effect, where the student teaches the AI acting as a naive tutee).* The outcomes measured in the study are *(B) learning performance (objective test scores), (C) self-efficacy (confidence in one's ability), and (D) confidence calibration (alignment between confidence and actual correctness).*

This study hypothesizes that reverse-tutoring will lead to better knowledge mastery, stronger self-efficacy, and better alignment between confidence and actual competence than both the Socratic chatbot and the direct-answer chatbot, as teaching forces learners to organise knowledge and expose gaps, reducing the illusion of competence while supporting deeper learning.

4 Methodology

4.1 Overview

This research plans to conduct a within-subjects study by utilizing the Latin Square Design to investigate the effects of different generative AI tutoring styles (direct-answer, Socratic, and reverse-tutoring). Participants will experience all three styles to help them learn data structures and have their knowledge tested across three phases: pre-test before learning with AI, practice-test with AI, and finally a post-test after the AI is taken away. In order to measure the overall effect of the different AI tutoring styles, the following 3 factors will be considered: the objective learning outcome based on their performance in the multiple-choice assessments, their self-efficacy measured through their predicted correct answers and 5-point Likert scale, as well as their confidence calibration using the Brier score. Finally, the collected data will be analyzed using Generalized Linear Mixed Models (GLMM) for continuous measures and Cumulative Link Mixed Models (CLMM) for ordinal psychological data.

4.2 Materials

4.2.1 Study topic and task

Data structures are chosen as the topic for this study [49]. There will be 3 sets of subtopics that will be used in the study:

1. Array sorting algorithms
2. Linked Lists
3. Binary Search Trees

Data structures are chosen as the topic for this study because they possess a good balance of high-level computational thinking, pattern recognition, and algorithm design. Since these topics very in difficulty, we will be counterbalancing the order of the tutoring styles and the topics to ensure the experience is balanced. Furthermore, data structures are a subject that is uncommon to the general public, making it a neutral and new topic for the participants, especially for those outside of the computer science field.

To ensure a self-paced learning experience, participants will be given a material video that explains the topic in detail at the start of the study [49]. Then, participants will be given a time limit of 30 minutes to study using the assigned AI. This time limit is chosen to ensure users are given enough time, and not feel too fatigued while undergoing all three different tutoring styles. Whenever they feel ready, they can freely start the assessment. The assessment consists of 3 sets of 5 multiple-choice questions each to measure the participant's knowledge on a topic for pre-test, practice-test with AI, and post-test when the AI is taken away [18]. Therefore, a total of 45 distinct questions will be answered by

the participant across the three stages. Furthermore, all participants will be doing the study in a lab setting to ensure a controlled environment and discourage them from cheating.

4.2.2 AI prompts and guardrails

In creating the interface that the participants will interact with, an AI chatting interface will be developed using Next.js and the Gemini SDK. Prompt engineering and guardrails will be used to design the personality of the AI to ensure the interaction matches the expected dialogue pattern [16]. The prototype will then be pilot tested internally to ensure it is working as expected. All three tutoring styles will be given the same temperature and configuration, ensuring consistent output accuracy and performance across all three tutoring styles [14]. Additionally, participants will be neutrally briefed on the type of AI tutoring style they have been assigned to, along with a guide on how to properly interact with it. This ensures the participants understand the expected dialogue pattern.

4.3 Participants

This study aims to recruit a minimum of 60 participants, obtained from power analysis with a combination of 0.25 effect size, 0.05 alpha, and 0.80 power, along with 5 predictors (2 from tutoring styles (n -1 rule), AI literacy, programming proficiency, and trust in AI). In maintaining a diverse sample, participants will be recruited from different backgrounds (i.e. gender and age) and across a spectrum of programming proficiency. Participants will be recruited through an expression of interest form, which will be shared across the university only after the ethics application is approved. They will then be given a selection of schedules that they can choose from to participate in the study. To ensure efficient use of time and resources, students will be scheduled in groups of five for each lab session.

4.4 Data Collection

Data will be collected through a series of surveys and questionnaires from onboarding to the end of the study. Participants' demographics, background, AI literacy, and level of trust in AI will be collected in the expression of interest during the recruitment process. Additionally, participants will be asked to fill out a self-report programming proficiency questionnaire to measure their experience. The questionnaires consist of 5-point Likert scale questions regarding their self-estimation of their programming experience and logical paradigm knowledge [50].

All participants' names will be replaced with a code number to ensure anonymity. Regarding the study itself, three sets of five multiple-choice questions will be used to assess the participants' objective learning performance across the different stages of the study (pre-test after the initial briefing, practice-test after learning with AI, and post-test with the AI taken away). In parallel, a questionnaire will be given after each assessment to measure participants' confidence levels in their self-efficacy and predicted correct answers. The data will be collected using a 5-point Likert scale to measure how confident they are feeling, and also an option from 1 to 5 for their predicted correct answers, where both data will be collected via Qualtrics. Since the participants will be participating in a within-subject study, they will be invited to take three sessions to experience all three tutoring styles. Therefore, this data collection process will be repeated three times, totaling nine sets of assessments and confidence questionnaires. Additionally, the duration that each participant spends during their learning sessions and assessments, as well as the chat logs, will be collected for analysis and comparison between different styles. All three tutoring styles will be evaluated in one lab session to avoid participants doing independent learning outside of the study. To avoid fatigue and burnout, participants will be given a break between each session. Furthermore, participants will be compensated hourly, and participant will receive a bonus based on their performance in the study to ensure they are motivated to perform their best.

4.5 Dependent and Independent Variables

Variables are defined consistently with the research question and hypotheses in section 3.

The **independent variable** for this study is the different *tutoring styles* that will be manipulated at three levels:

1. direct-answer tutoring as the control,
2. Socratic guidance,
3. reverse tutoring (learning-by-teaching).

Each participant will experience all three styles in a Latin square randomization order to ensure a balanced experience among all three styles. Moreover, the study is conducted in a within-subjects design to observe how the same participant performs across different tutoring styles and to achieve stronger statistical power.

On the other hand, the **dependent variables** are:

- *Learning performance.* Objective scores retrieved from the multiple-choice assessments, across the pre-test, practice-test with AI, and post-test without AI.
- *Self-efficacy.* Scores from a self-efficacy questionnaire given to the user after every assessment, capturing confidence on a 5-point Likert scale, and the number of predicted correct answers.
- *Confidence calibration.* Alignment between self-rated confidence and objective correctness. This is computed from the item-level confidence ratings and actual learning performance.

The **covariates** included in this study are the participants' previous programming proficiency, AI literacy, and their propensity towards AI, which are drawn from previous studies [18, 13, 11]. To capture the participants' AI literacy, the Meta AI Literacy Scale (MAILS) [51] modular framework will be used. This framework consists of a number of questionnaires with answers on a scale of 1 to 10. Then, to capture the participants' trust in and propensity toward AI, the Trust in Automation (TiA) questionnaire will be used, which is a widely used set of questionnaires in the human-computer interaction field to measure the dynamics between humans and automated systems [52]. Alongside these, the participants' previous programming proficiency will be collected through a self-report questionnaire [50].

4.6 Data Analysis

After all the studies have been conducted, the raw data collected from the platform logs, assessments, and questionnaires will be processed to construct the primary dependent variables. The objective learning performance will be calculated based on the sum of the correct answers on the 5-item multiple-choice questions. Then, interaction time will be extracted from the platform logs to measure the duration each participant spent interacting with the AI to study the topic before proceeding to complete the assessment. In order to measure the confidence calibration, calculation of the Brier Score [53] will be used for each participant. This method is a proper scoring rule which is used to measure the accuracy of probabilistic prediction to quantify the gap between perceived confidence and the absolute outcome [53]. In this study, the Brier Score will be calculated by comparing the participant's predicted correct answers with the actual correct answers.

The Brier score (BS) for confidence calibration is formally defined as:

$$BS = \frac{1}{N}\sum_{t=1}^{N}(f_t - o_t)^2 \tag{1}$$

where f_t represents participants' predicted correct answer count out of 5, and o_t represents the actual correct answers. In order to equally measure the confidence calibration, we will be utilizing the Brier score. A student who predicts 4/5 and scores 2/5 (overconfident) will receive the same score as a student who predicts 2/5 and scores 4/5 (underconfident). Hence, the magnitude of the miscalibration is measured as opposed to the direction.

Since our study employs a within-subjects Latin Square design, each participant will experience all three AI tutoring styles. As a result, their data points are intrinsically linked; for instance, a quick learner may exhibit consistently high scores across styles due to their baseline ability rather than the influence of the AI tutoring styles.

To address this dependency, we utilize mixed-effects models over traditional statistical tests, such as standard ANOVAs, which assume that each data point came from an independent individual. In contrast, mixed-effects models distinguish between within-participant factors (such as AI tutoring style) and between-participant variability (such as individual ability). Including the participant's ID as a random effect allows the model to account for the differences in each participant's cognitive ability. Hence, it allows us to accurately isolate the impact of the AI's tutoring style itself [54].

Our analysis accommodates two primary data types:

1. **Generalized Linear Mixed Models (GLMM):** Applied to objective, continuous measures, including learning performance (scores out of 5), interaction time (minutes studying), and Brier Score (confidence calibration).

2. **Cumulative Link Mixed Models (CLMM):** Used for ordinal psychological data, specifically self-efficacy scores from 5-point Likert scale questionnaires. These models account for the ranked, but not interval, nature of such responses [55].

In order to control for confounding variables, participants' previous programming proficiency, AI Literacy (MAILS) and Trust in Automation (TiA) scores are included as covariates in all models. Additionally, the logs will also be analyzed using thematic analysis to identify any key themes and patterns in participants' interactions with the AI, such as frustrations or positive interactions.

Finally, in determining which AI tutoring style is most effective, we first will check to see if the differences in tutoring styles produced a significantly different outcome using the mixed-effects models. After that, we will then conduct a post-hoc pairwise comparison using Estimated Marginal Means (EMMs) with Bonferroni correction. By doing so, it allows us to compare the different styles head-to-head while mitigating false positives. The most effective style will be identified holistically based on the interaction of our metrics of being able to achieve the highest learning performance while maintaining high self-efficacy and producing a low Brier score.

For each outcome (test score, study time, Brier score, or Likert self-efficacy via a CLMM), we fit the same structure: the tutoring style in that session, the participant's MAILS, TiA, and self-reported programming proficiency, with a random intercept per person since everyone will be completing all three styles.

$$Y_{ij} = \beta_0 + \beta_1 \,\mathrm{Style}_j + \beta_2 \,\mathrm{Literacy}_i + \beta_3 \,\mathrm{TiA}_i + \beta_4 \,\mathrm{ProgExp}_i + u_i + \varepsilon_{ij}, \tag{2}$$

Here i represents the index for participants and j represents the index for their Latin-square sessions. Then β_0 is the intercept, β_1 captures tutoring style (direct-answer, Socratic, reverse tutoring), β_2–β_4 capture the three covariates, u_i is the person-specific baseline shift, and ε_{ij} is the error left over for that person-session.

4.7 Ethics

This study commits to upholding the principles of conducting research involving humans by following ethical guidelines[56]. Importantly, the study will commence only after the ethics application is approved by the University of Melbourne. In following ethical principles, safety of the participants will be respected and ensured, breaks are provided in between sessions to avoid any discomfort. All participants are also voluntary and free from any coercion. Before the study begins, participants will be given a consent form with a plain language explanation of the study's description, goals, risks, and expected time commitment. Additionally, participants will also be compensated for their time and effort. In order to protect participants' privacy, all names will be anonymized and their data will be stored securely in a password-protected database. Finally, participants are free to withdraw from the study at any during the study time without penalty, with their data being erased from the database.

4.8 Timeline

Figure 1 presents the planned schedule for this study across Semester 1 and Semester 2 of 2026.

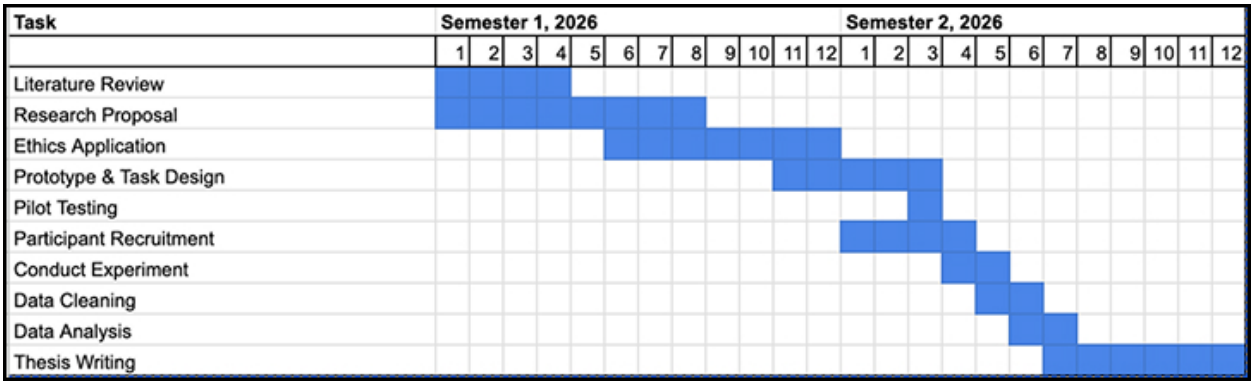

Figure 1: Project timeline and milestone plan for the proposed study across Semester 1 and Semester 2 of 2026 (by week).

5 Conclusion and contribution

The goal of this study is to investigate the effectiveness of learning using AI with different tutoring styles. This study examines effectiveness by measuring participants' self-calibrated confidence, actual objective learning performance, and the confidence calibration between the two. We plan to uncover the illusion of competence [18] by using the learning-by-teaching AI tutoring style to see whether the protégé effect could mitigate the illusion and, in parallel, positively impact the user's self-efficacy and learning outcome. This is done by conducting a within-subject study across three different tutoring styles (direct-answer, Socratic, and learning-by-teaching), where students are tasked to learn data structures using the assigned AI to be assessed with and without the AI. By doing so, we will be able to analyze which tutoring style best supports self-efficacy and learning outcomes.

This research contributes to the field of Human-AI interaction by addressing the research gap in the design of human-centric AI learning tools [5]. As found in previous studies, especially on the exploration of Socratic chatbots, the design of these chatbots is able to increase users' learning outcomes [38, 21]. However, beyond Socratic chatbots, there is a lack of research on the design of other AI tutoring styles. By introducing the learning-by-teaching AI, we are able to explore a new interaction design and also compare the effectiveness of another AI learning tool. Furthermore, this study seeks to evaluate whether the protégé effect could be translated into human-AI interaction from the traditional human-to-human medium. Ultimately, future AI learning platforms will be able to utilize our research findings to design more effective and human-centric AI learning tools.

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